

Executive Summary

This lab focused on implementing defense-in-depth network security using Azure Virtual Networks, Application Security Groups (ASGs), and Network Security Groups (NSGs). The goal was to create a secure environment where web servers are publicly accessible only for HTTP/HTTPS traffic, while management servers are restricted to RDP access from authorized sources.

I successfully achieved this by:

- Creating a VNet and subnet
- Defining two ASGs (web and management)
- Configuring NSG rules with ASG-based filtering
- Deploying two test VMs and validating access control

The lab demonstrated least-privilege networking, scalable rule management, and practical troubleshooting.

Task 1: Create Virtual Network and Subnet

- Created a virtual network named **FuturinCLOUD-VNet** with address space 10.0.0.0/16.
- Added a subnet named default (10.0.0.0/24).
- This provided the foundation for all VMs and security rules.

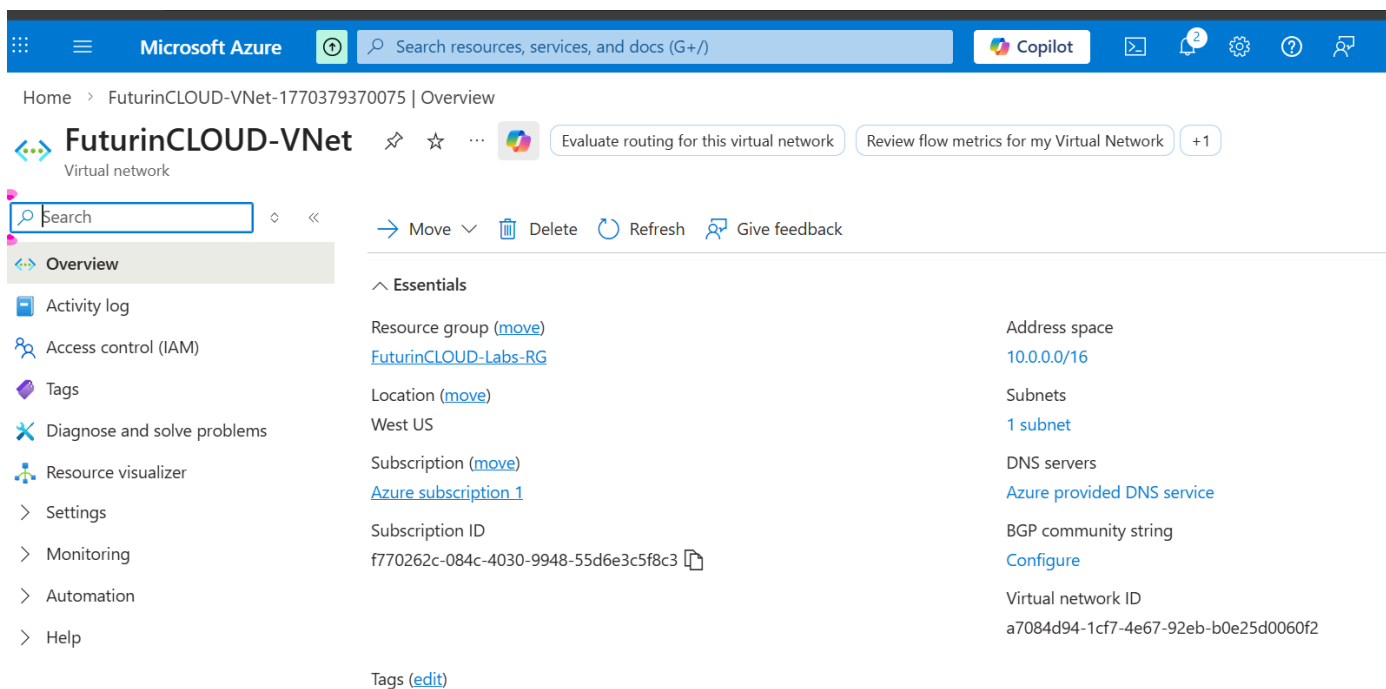


Figure 1 – Virtual network and subnet configuration

Task 2: Create Application Security Groups (ASGs)

- Created myAsgWebServers for web-facing VMs
- Created myAsgMgmtServers for management/admin VMs
- ASGs act as tags, allowing scalable, group-based rules instead of IP-based ones.

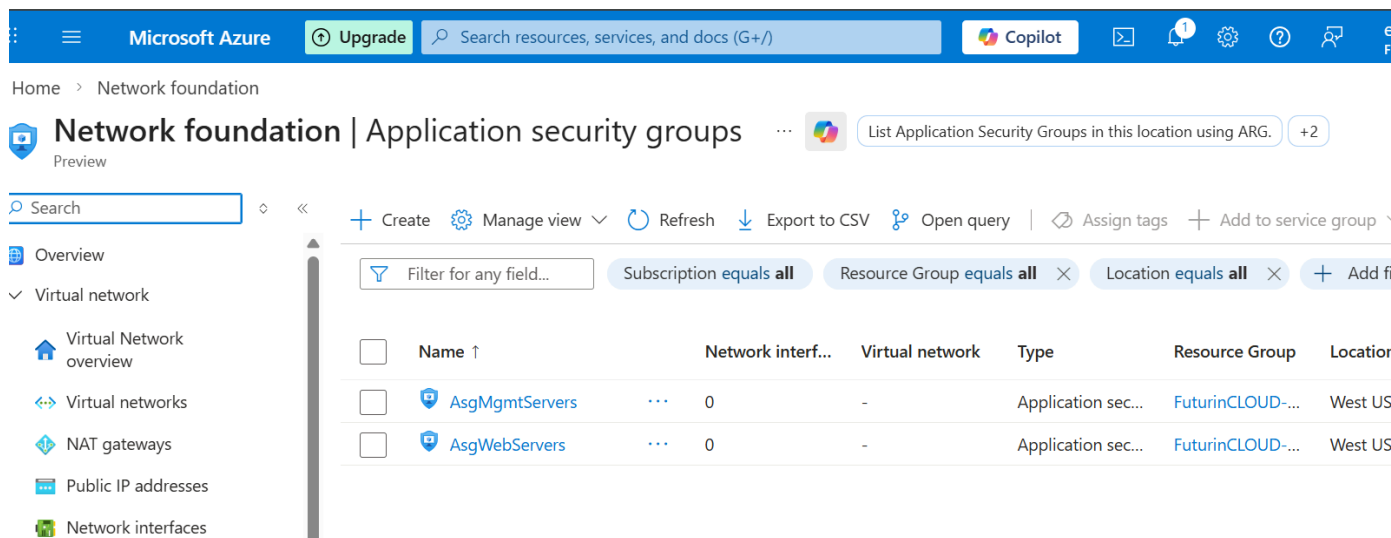


Figure 2 – Application Security Groups created for logical server grouping

Task 3: Create and Configure Network Security Group (NSG)

- Created NSG named myNsg (or FuturinCLOUD-NSG)
- Associate the NSG with the default subnet
- Added inbound rules:
 - Allow HTTP (80) and HTTPS (443) from Any → destination myAsgWebServers
 - Allow RDP (3389) from Any → destination myAsgMgmtServers
- All other traffic is denied by default (implicit deny rule)

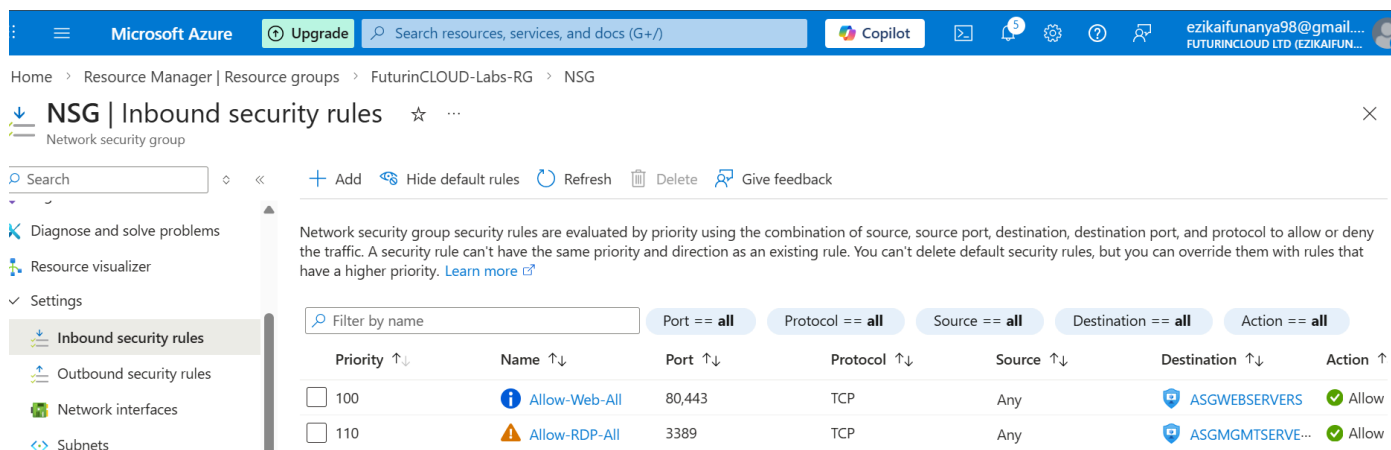


Figure 3 – NSG inbound rules enforcing least-privilege access

Task 4: Deploy Test VMs and Attach ASGs

- Deployed myVmWeb (web server) → attached to myAsgWebServers
- Deployed myVmMgmt (management server) → attached to myAsgMgmtServers
- Installed IIS on myVmWeb using Run command (PowerShell: Install-WindowsFeature Web-Server)

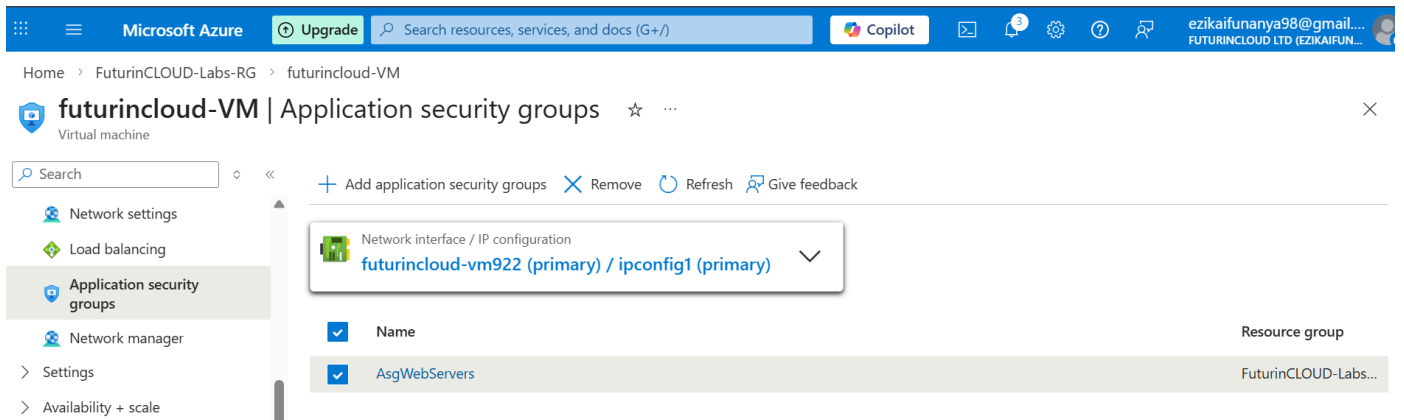


Figure 5: Attached VM to AsgWebservers

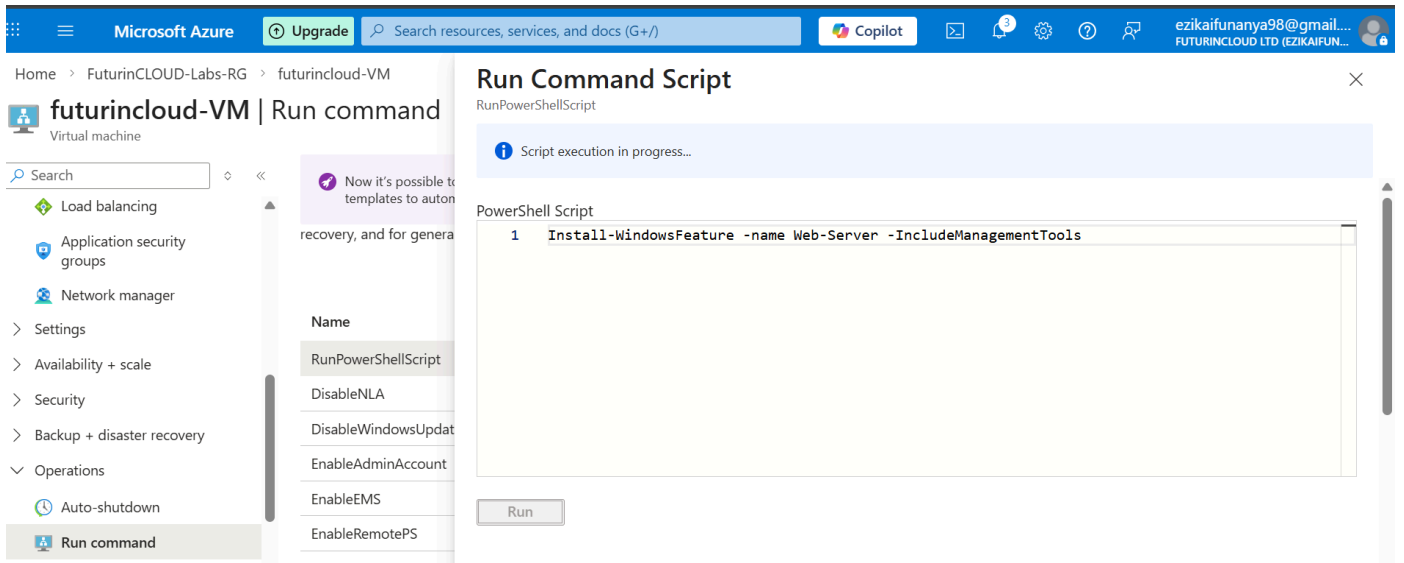


Figure 6: Installed IIS on myVmWeb using Run command

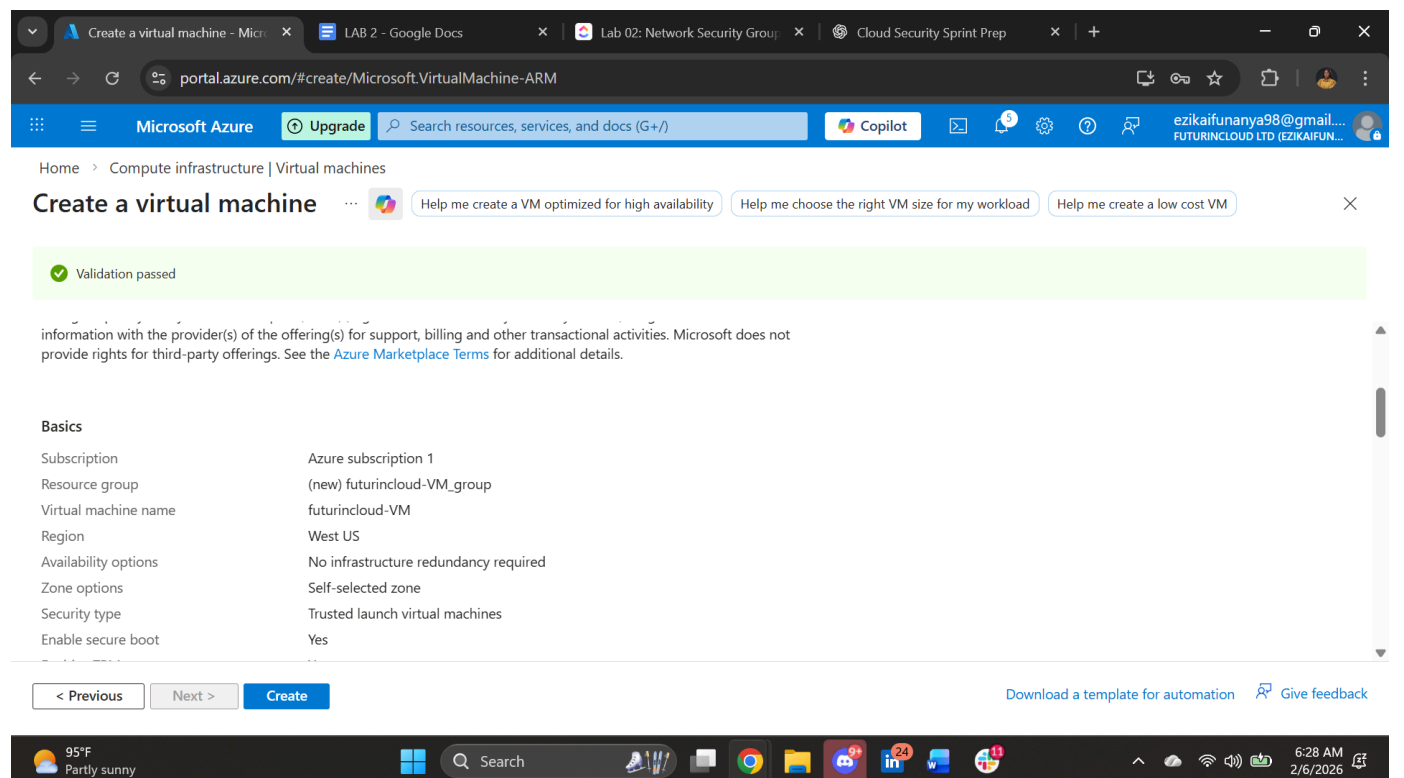


Figure 7: Created a virtual Machine

Task 5: Test and Validate Access Control

- From local browser → [http://\[myVmWeb public IP\]](http://[myVmWeb public IP]) → IIS welcome page loaded (success)
- Tried RDP to myVmWeb public IP → connection timed out / refused (blocked – success)
- RDP to myVmMgmt public IP → connected successfully (allowed – success)
- Confirmed NSG rules enforced segmentation correctly

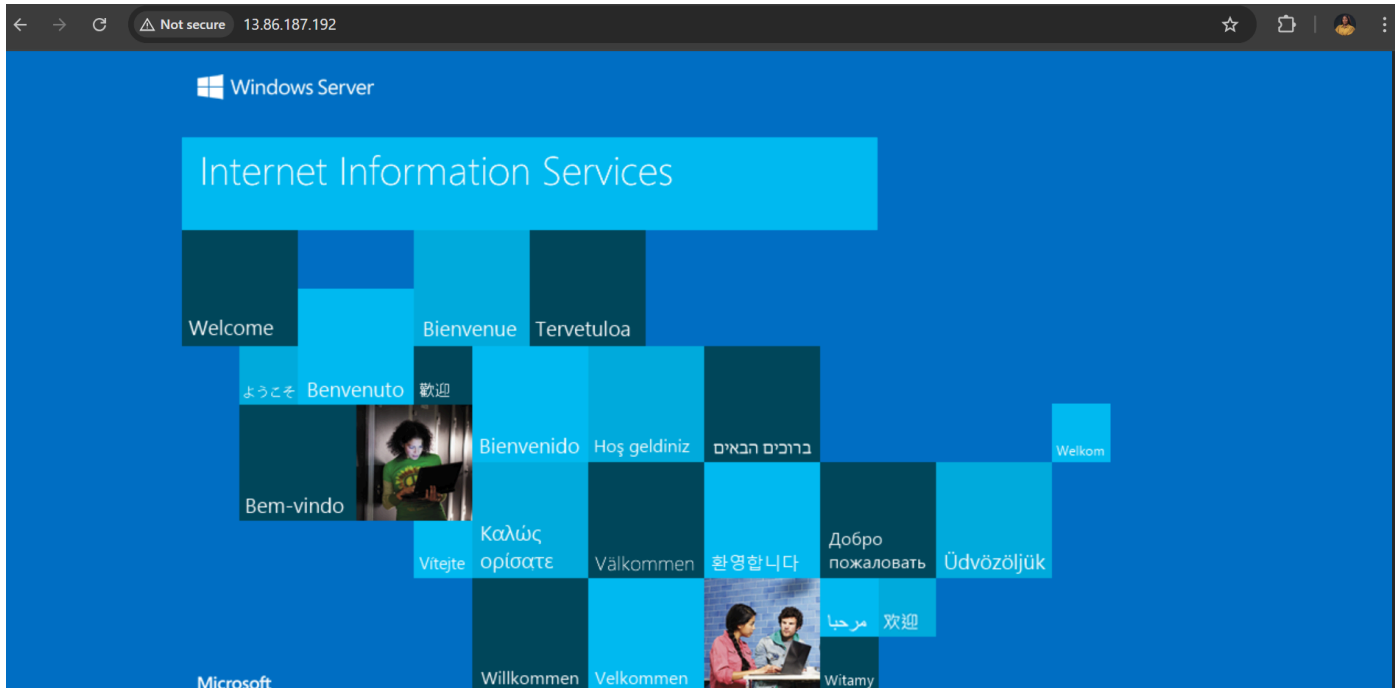


Figure 7 – Successful public HTTP access to web server

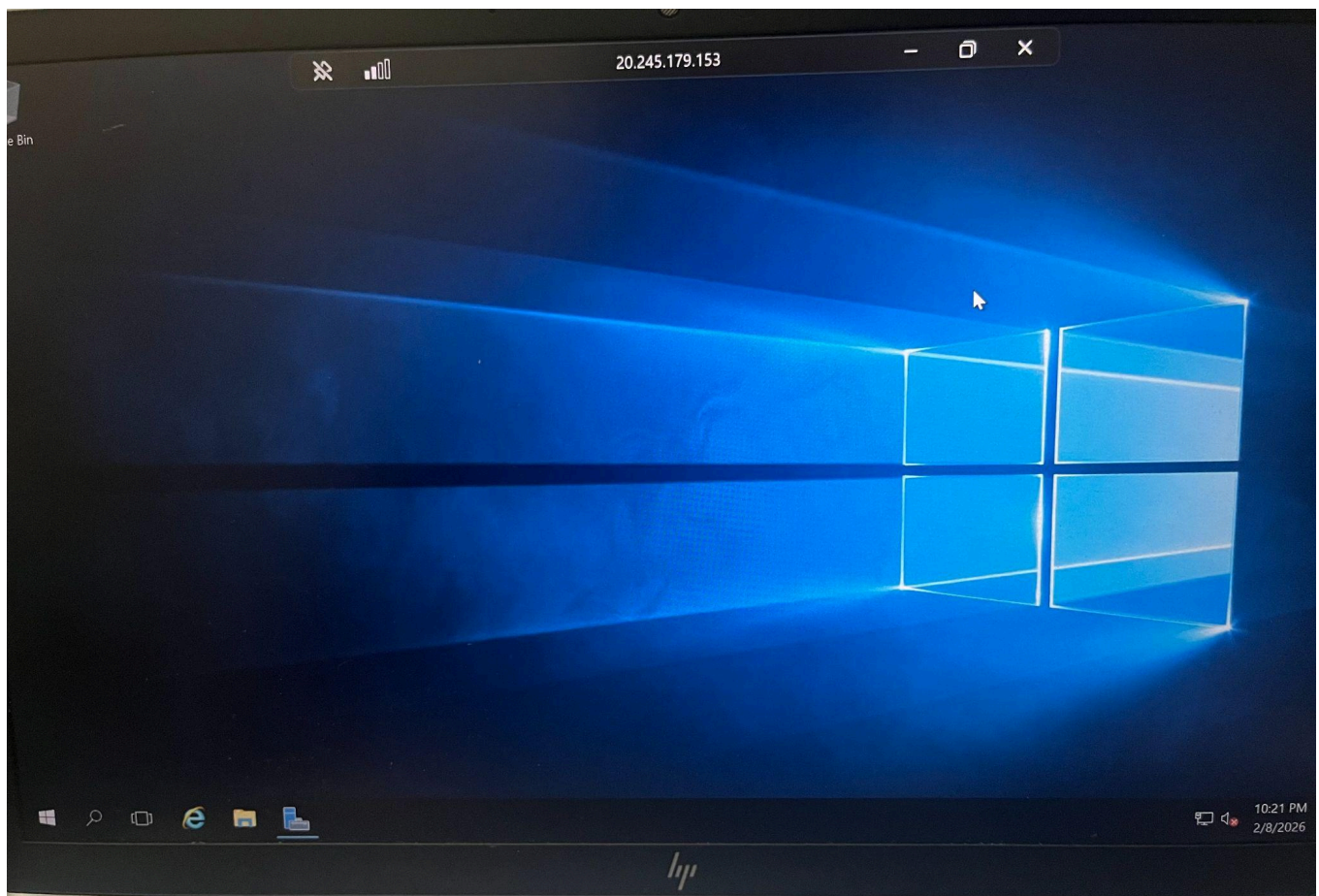


Figure 8 – RDP allowed to the management server

Recommendations

The following recommendations are proposed to enhance and extend the network security posture in a real-world or production-like FuturinCLOUD environment:

- Implement Network Watcher for traffic analytics, packet capture, and NSG flow logs to gain visibility into actual traffic patterns and detect misconfigurations or anomalies.
- Enable Azure DDoS Protection Standard on the VNet to protect public-facing web workloads against volumetric and protocol attacks.
- Use Azure Firewall (or Azure Firewall Premium) as the next layer for centralized egress filtering, application-layer inspection, and threat intelligence.
- Adopt Azure Policy to enforce NSG and ASG standards across all subscriptions and resource groups (e.g., require ASGs on every NIC, deny public IPs on management VMs).
- Integrate Microsoft Defender for Cloud to continuously assess NSG rules, surface risky configurations, and recommend hardening steps.
- Create custom RBAC roles that combine VM Contributor with specific NSG management permissions, reducing the blast radius of administrative accounts.
- Document and automate NSG/ASG deployments using Bicep or Terraform templates to ensure consistency and repeatability in future environments.

Conclusion

Lab 2 successfully demonstrated the implementation of **secure network segmentation** in Azure using Virtual Networks, Application Security Groups (ASGs), and Network Security Groups (NSGs).

Key accomplishments include:

- Creation of a VNet with a properly sized subnet
- Logical grouping of workloads through two ASGs (web and management tiers)
- Configuration of targeted inbound NSG rules enforcing least-privilege access
- Deployment and correct tagging of two test VMs
- Practical validation confirming that HTTP/HTTPS traffic reaches only the web tier while RDP is restricted exclusively to the management tier

This configuration effectively prevents lateral movement, reduces the attack surface, and aligns with zero-trust networking principles. The use of ASGs instead of IP-based rules provides scalability and maintainability — a critical advantage in dynamic cloud environments. The lab clearly illustrates that proper network security controls are foundational to protecting cloud workloads

Final takeaway: Network segmentation using ASGs and NSGs is a simple yet powerful technique that delivers strong security outcomes with minimal operational complexity — a core skill for any Azure security practitioner.